GNSS Water Vapour Remote Sensing

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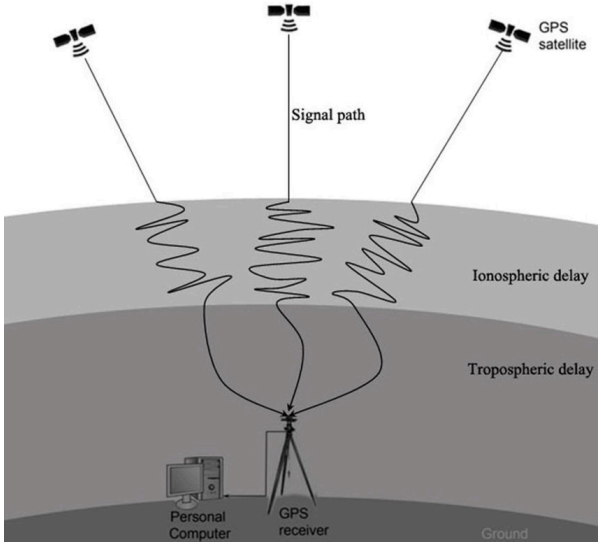
GNSS (Global Navigation Satellite System)

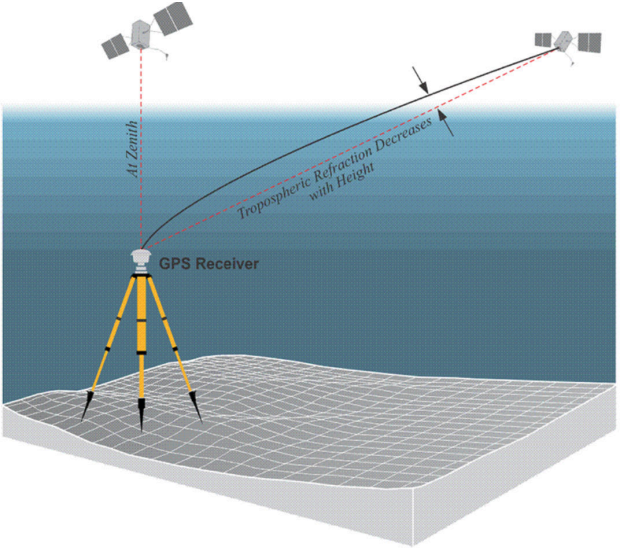
GNSS water vapour monitoring is a powerful technique that turns geodetic, ground-based GPS receivers into atmospheric weather stations. It works by measuring how much water vapour delays satellite signals traveling through the atmosphere.

The Basic Principle:
As GNSS signals (GPS, Galileo, etc.) travel from satellites to a ground station, they pass through the troposphere, the lowest layer of Earth's atmosphere. Water vapour is a "polar molecule" (it has a positive and negative end) and significantly refracts (slows down) the radio waves.

Simplified GNSS Delay Principle



Simplified GNSS Delay Principle



GNSS Delay Principle

Signals from GNSS satellites (GPS, Galileo, GLONASS, BeiDou) travel from space to a ground receiver at nearly the speed of light. When they pass through the neutral atmosphere (troposphere), they are slowed down and slightly bent due to refraction.

This delay is called the tropospheric delay and contains information about:

- Air pressure
- Temperature
- Water vapor content

Because water vapor strongly affects radio signal propagation, GNSS can be used as a passive atmospheric sensor to estimate atmospheric water vapor.

Units

Note that Integrated Water Vapor (IWV) is also commonly referred to as Total Column Water Vapor (TCWV) and Precipitable Water (PW) or Precipitable Water Vapor (PWV).

TCWV and IWV are typically expressed in units of kg/m^2, while PW and PWV represent the equivalent water height, usually given in cm or mm.

These units are interchangeable, as 1 kg/m^2 of IWV corresponds to 1 mm of PW.

GNSS Delay Principle

Total Tropospheric Delay

The total delay in the zenith direction is called:

$$ZTD = ZHD + ZWD$$

Where:

ZTD = Zenith Total Delay
ZHD = Zenith Hydrostatic Delay (dry component)
ZWD = Zenith Wet Delay (water vapor component)

GNSS Delay Principle

ZHD = Zenith Hydrostatic Delay (dry component)

- Caused by dry gases (mainly nitrogen and oxygen)
- Accounts for ~90% of total delay
- Can be modeled very accurately from surface pressure or reanalysis data
- Stable and predictable

GNSS Delay Principle

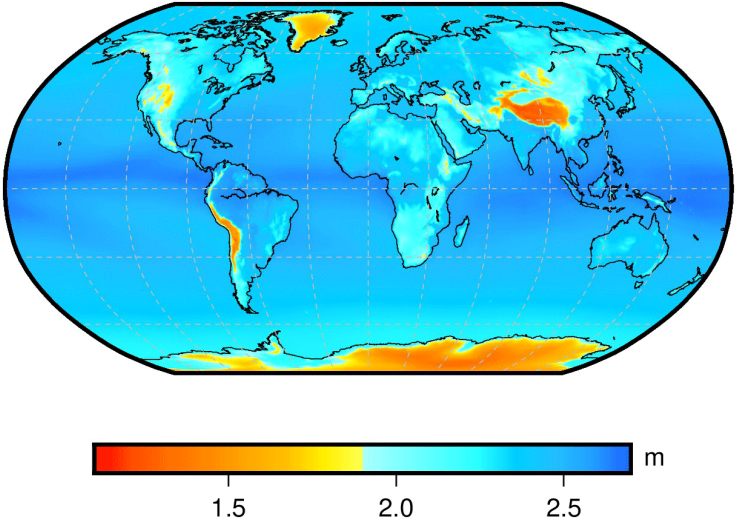
Using the Saastamonien model:

$$ZWD = ZTD - ZHD$$

$$ZHD = \left[\frac{2.2779 \pm 0.0024}{1 - 0.00266\cos(2\phi) - 0.00028h}\right] P_s$$

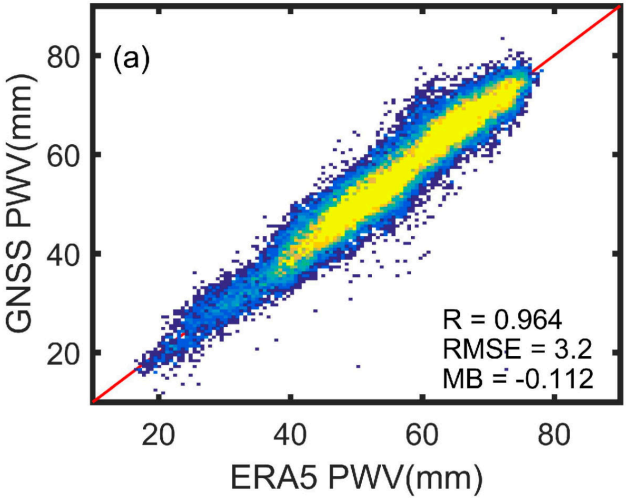
P_s is the ground pressure at the station (unit: hPa).
h is the height of the station (unit: km).
ϕ is the latitude of the station.

Zenith Total Delay from ERA5 (2015)



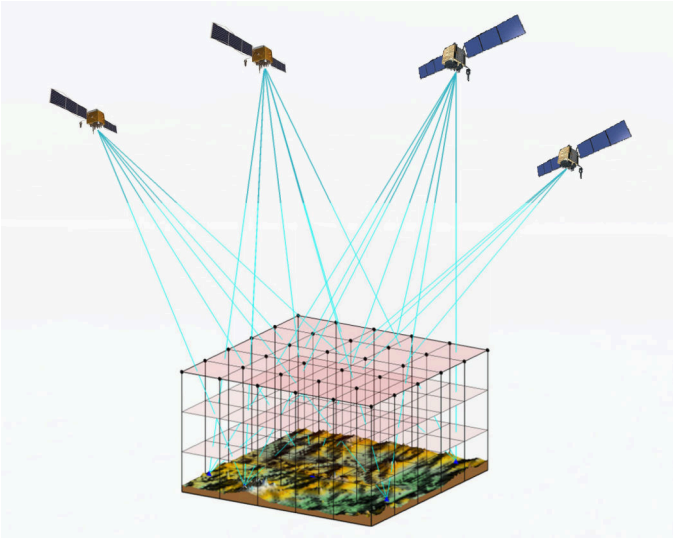
Huang et al., 2023

Correlation of GNSS-IWV and ERA5-IWW



Su et al., 2025

GNSS Water Vapour Voxel principle



Z. Dong and S. Jin, 2018

Station NPA1 (489 m)

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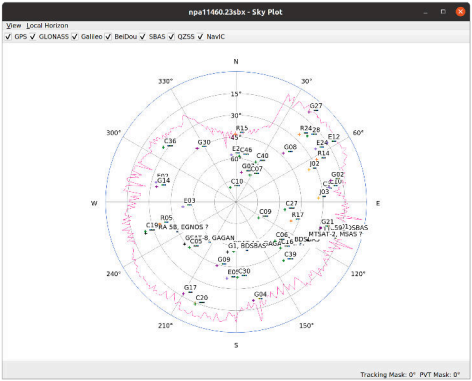
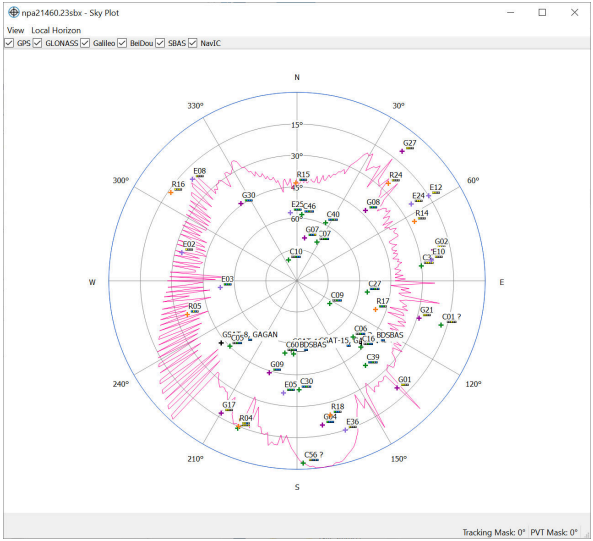
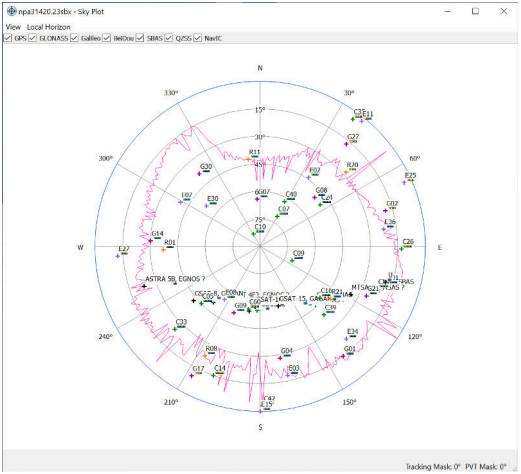


Station NPA2 (840 m)

Station NPA2 (840 m)



Station NPA3 (1700 m)



How to access pre-processed data?

Use the pre-processed data at the Nevada Geodetic Laboratory (NGL):
https://geodesy.unr.edu/NGLStationPages/stations/KUGE.sta

Download RINEX KUGE data from the UNAVCO website:
https://www.unavco.org/data/doi/10.7283/FXPT-T160

Use the CSRS-PPP from the Canadian Government (requires registration, but is free):
https://natural-resources.canada.ca/maps-tools-publications

Precise processing will require a GNSS-processing setup that is also solving for station-position uncertainties. The commercial software Bernese and the software package GAMIT provides meteorological observation and ZWD results.

Next Steps

We are going to download and use the pre-processed delay data from the NGL and plot the time series with matplotlib (Python).

Open the Jupyter Lab
`GNSS_first_steps_ZTD_KUGE_from_NGL.ipynb`